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(54) **REGULATION OF DNA SYNTHESIS BY NUCLEOTIDES LINKED TO PROTECTING GROUPS**

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C12Q 1/6834 (2018.01)

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(58) **Field of Classification Search**
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See application file for complete search history.

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(57) **ABSTRACT**

A method for DNA synthesis using protected nucleosides is disclosed. The nucleosides may be nucleoside triphosphates or nucleoside phosphoramidites with nucleobases attached to electrochemically-cleavable linkers. Removal of a protecting group by application of a voltage in solution triggers a cyclization reaction that cleaves the electrochemically-cleavable linkers. The electrochemically-cleavable linkers may include an amide linkage and an amide that forms a lactam or an ester linkage and a protected alcohol that forms a lactone when the protecting group is removed. The voltage used to cleave the electrochemically-cleavable linkers may be generated by activation of individual electrodes on a microelectrode array. The microelectrode array can be a substrate for solid-phase synthesis of oligonucleotides. Activation of specific electrodes removes the protecting groups at those electrodes and thus enables spatially-controlled extension of the oligonucleotides. Protected nucleosides linked to protecting groups by electrochemically-cleavable linkers are also disclosed.

20 Claims, 6 Drawing Sheets

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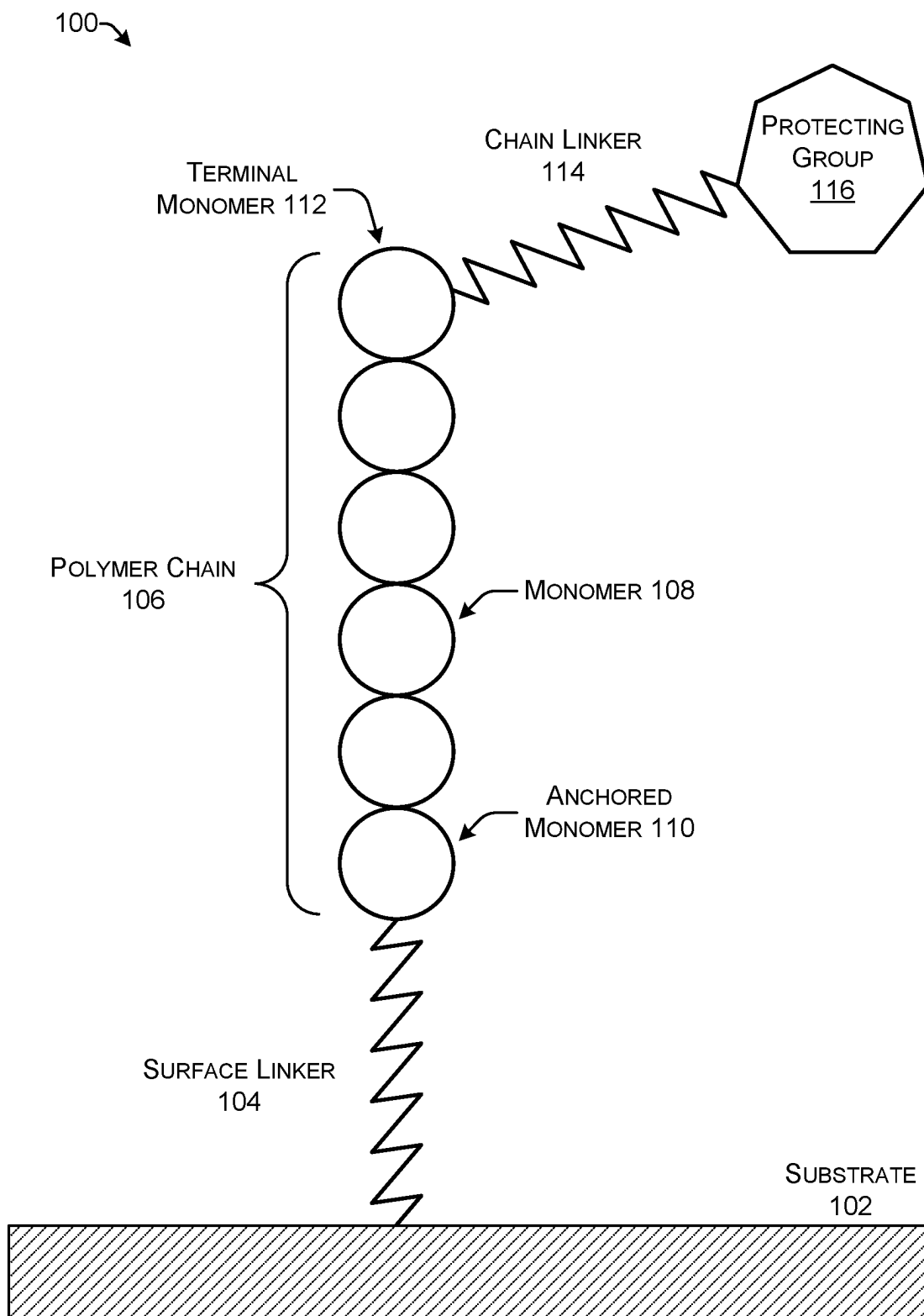


FIG. 1

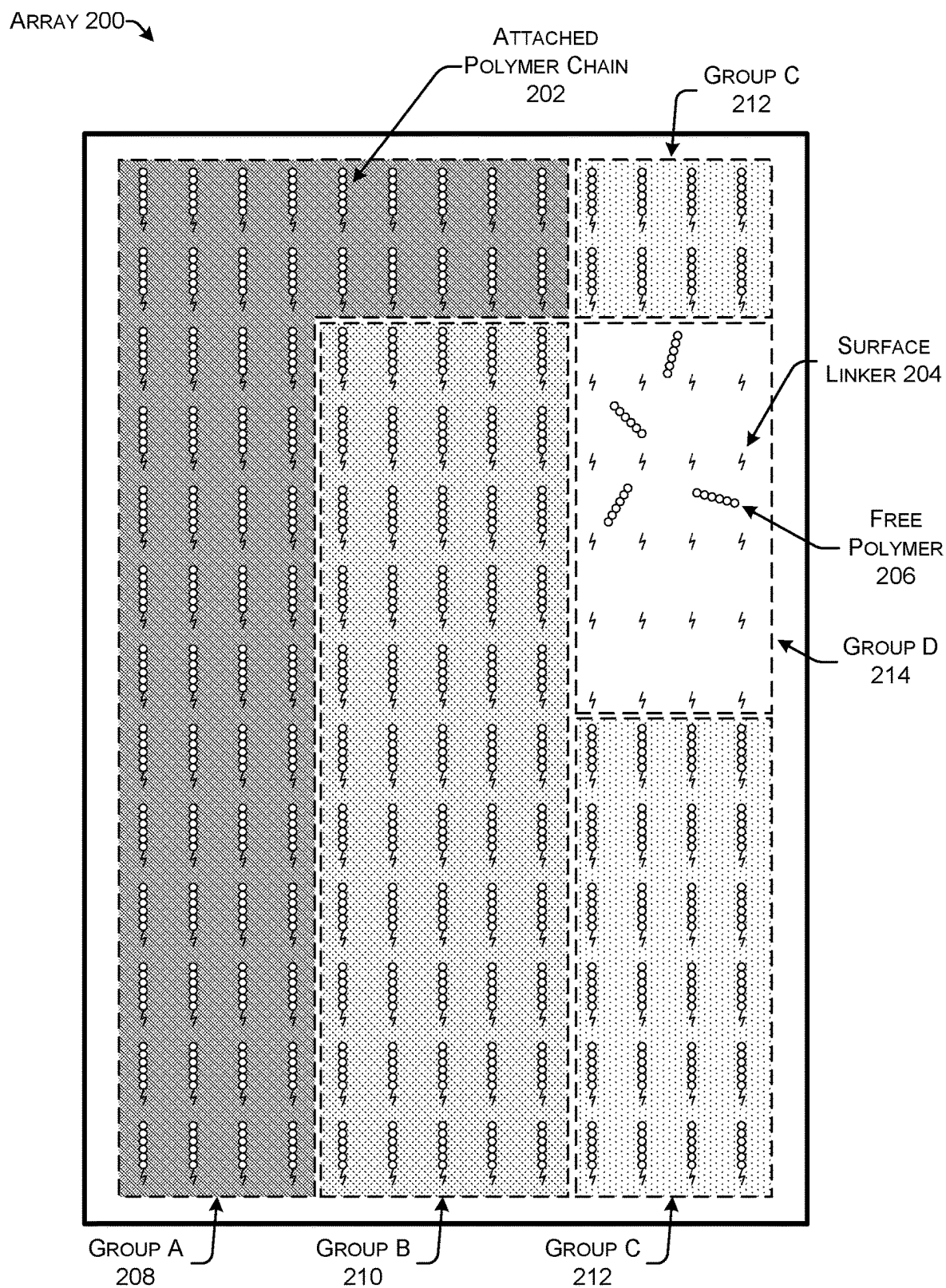
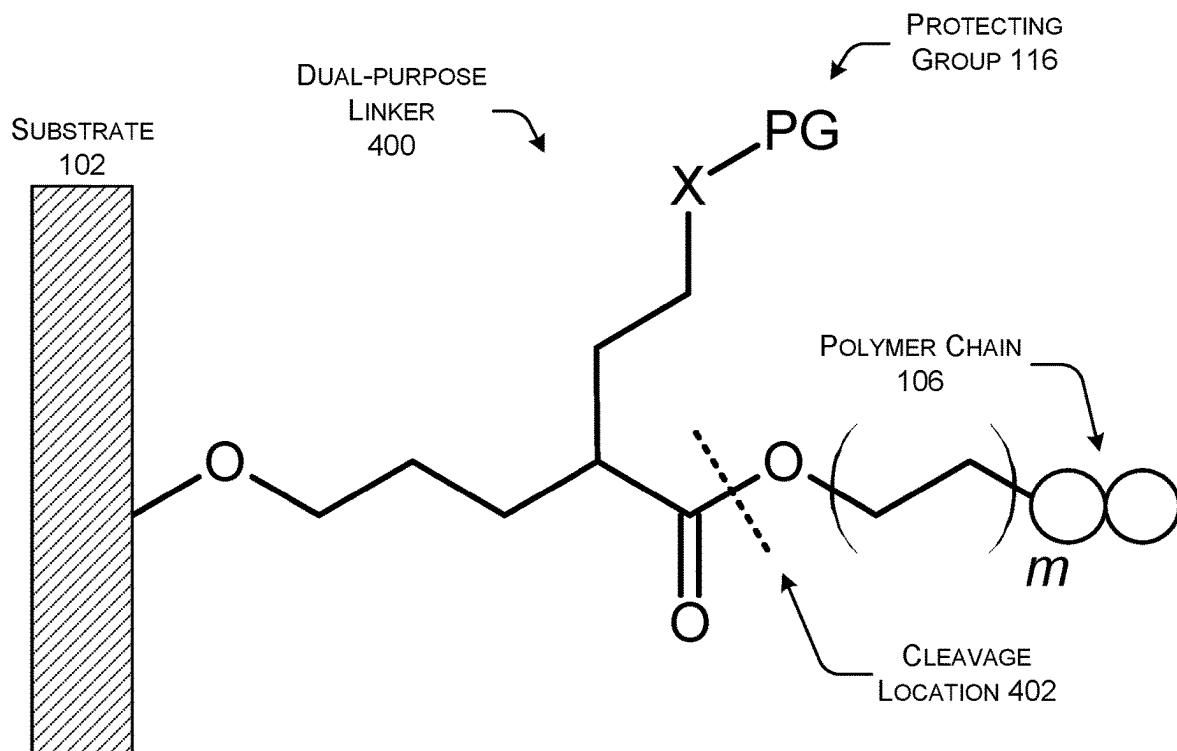
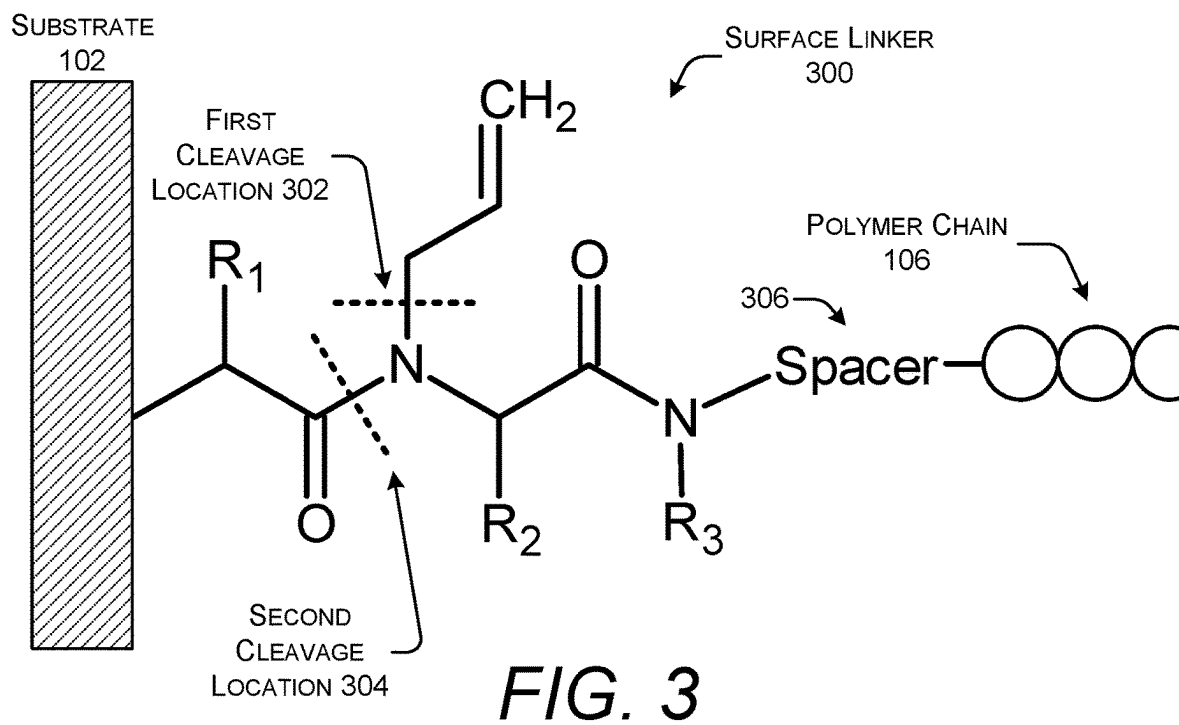
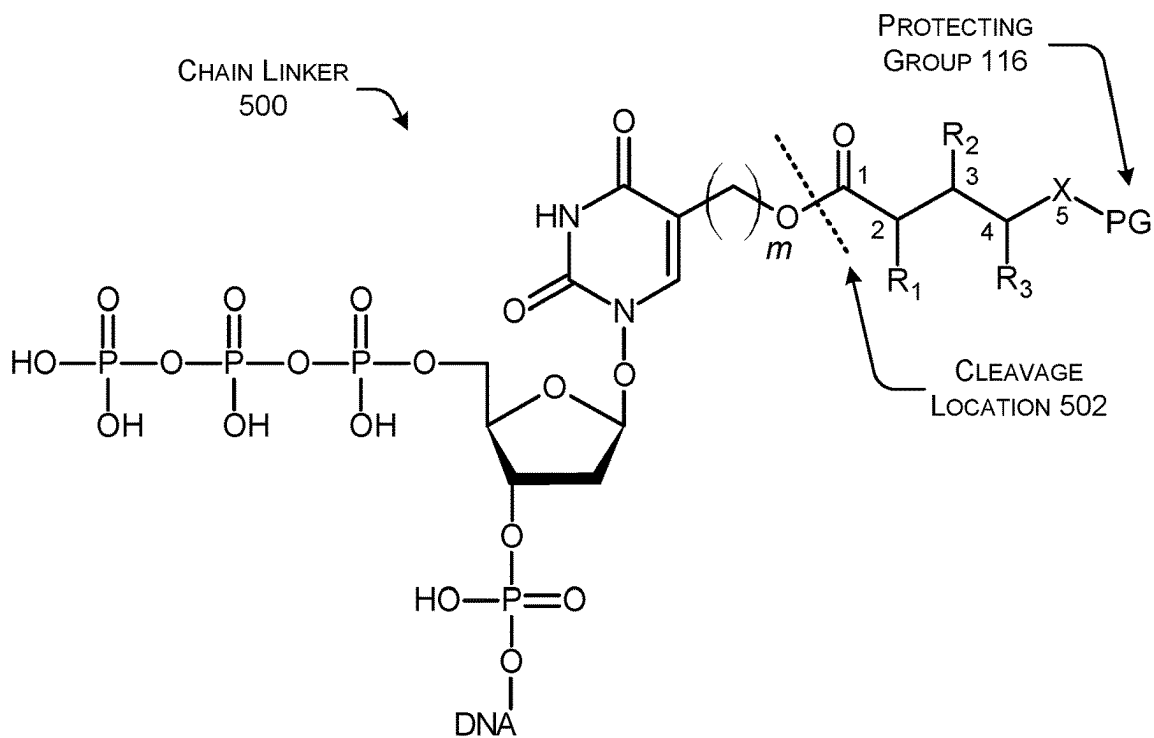
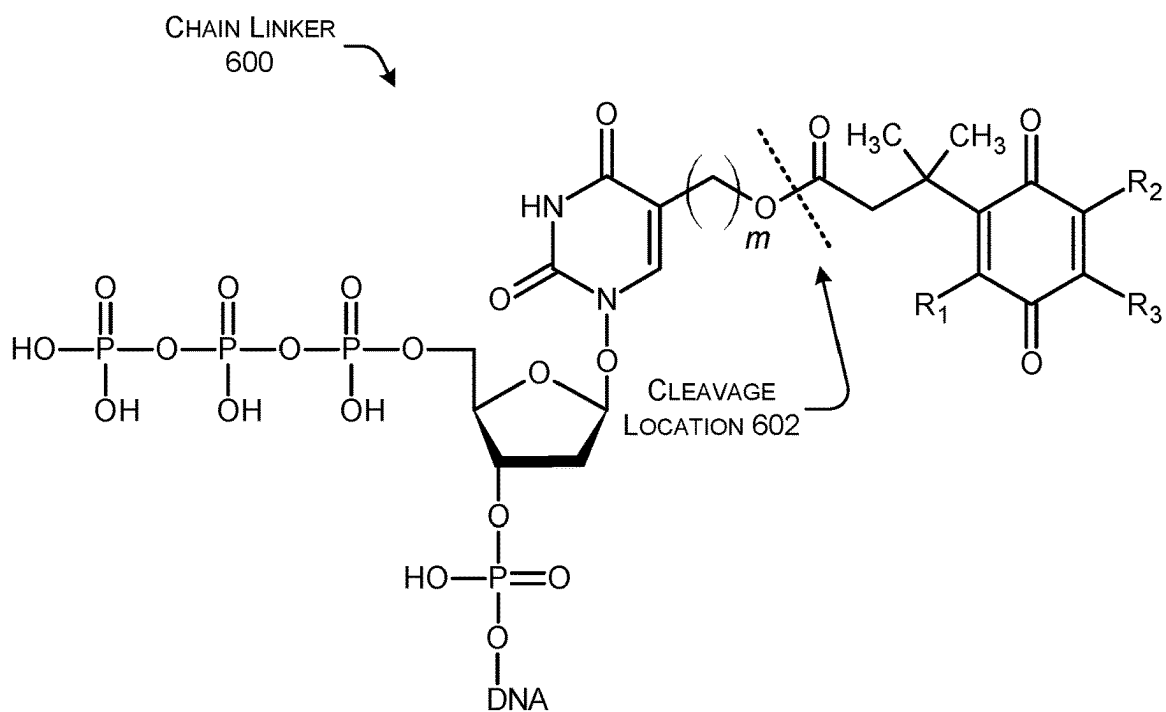
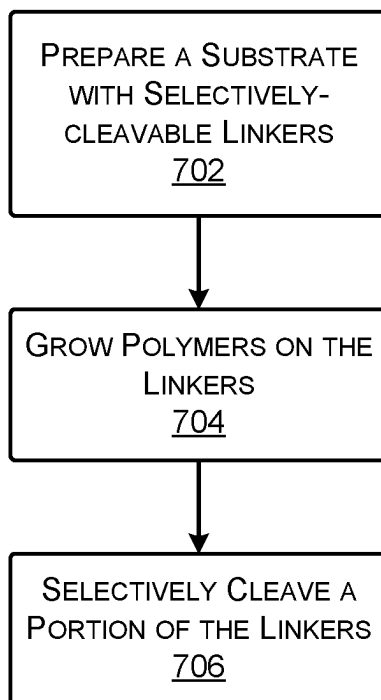


FIG. 2

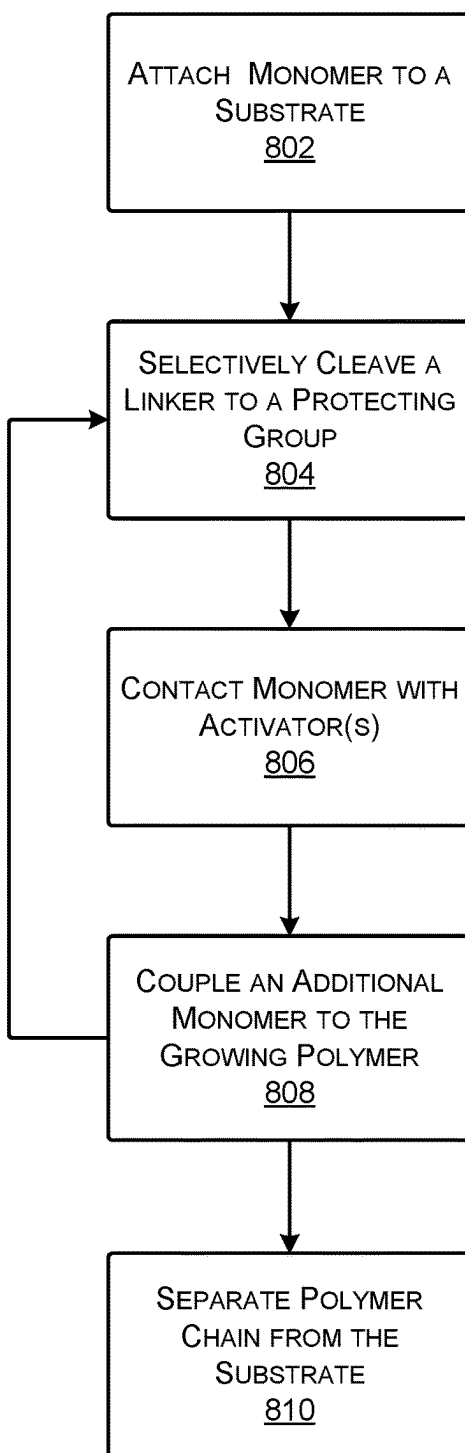


**FIG. 5****FIG. 6**

700 ↘

*FIG. 7*

800

*FIG. 8*

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REGULATION OF DNA SYNTHESIS BY NUCLEOTIDES LINKED TO PROTECTING GROUPS

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation-in-part of U.S. patent application Ser. No. 16/230,787 entitled “Selectively Controllable Cleavable Linkers” and filed on Dec. 21, 2018. The entire contents of which are expressly incorporated herein by reference in their entirety.

BACKGROUND

Modified nucleotides with protecting or blocking groups have long been used in the synthesis of oligonucleotides such as deoxyribonucleic acid (DNA) and ribonucleic acid (RNA). Lincoln J. McBride et al., *Nucleotide chemistry*, 16. *Amidine protecting groups for oligonucleotide synthesis*, J. of the Am. Chem. Soc., 1986, 108 (8), 2040-2048. One use for protecting groups is control of oligonucleotide extension. Protecting groups can be used to prevent addition of nucleotides. Removal of the protecting groups allows extension of a growing oligonucleotide chain. The protecting groups are typically removed using chemicals such as acids and bases.

Currently most oligonucleotides are synthesized by the phosphoramidite method developed decades ago as described in U.S. Pat. Nos. 4,415,732 and 4,458,066. Enzymatic DNA synthesis is a more recent technique that uses enzymes such as terminal deoxynucleotidyl transferase (TdT) rather than chemical synthesis for de novo manufacture of DNA strands. However, DNA polymerases such as TdT add nucleoside triphosphates to a growing oligonucleotide strand in an unregulated manner. One technique to force single-nucleotide addition uses protecting groups on the nucleotide triphosphates that block extension of the oligonucleotide strands. Examples of this technique are provided in Sebastian Palluk et al., *De novo DNA synthesis using polymerase-nucleotide conjugates*, Nat. Biotechnol. 36, 645-650 (2018) and U.S. Patent Application Serial No. 2019/0112627.

Linker structures and techniques for cleaving of the linkers, particularly without use of chemicals, can have utility in oligonucleotide synthesis both by the phosphoramidite method and the enzymatic method. This disclosure is made with respect to these and other considerations.

SUMMARY

This invention relates to modified nucleotides and their use in oligonucleotide synthesis. In particular, this invention discloses nucleotides having a removable protecting group, their use in oligonucleotide synthesis, and a method for electrochemical cleavage of the protecting group. In one implementation, this disclosure provides a method of controlling addition of nucleotides during oligonucleotide synthesis by activating electrodes in a microelectrode array to selectively remove protecting groups. In one implementation, this disclosure provides a nucleoside with a nucleobase linked to a protecting group by an electrochemically-cleavable linker, wherein the electrochemically-cleavable linker comprises (i) an ester linkage and a protected alcohol that forms a lactone when the protecting group is removed or (ii) an amide linkage and amine that forms a lactam when the protecting group is removed.

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This disclosure provides multiple techniques and linker designs for achieving selective cleavage of linkers. Selective control of linker cleavage allows for less than all the linkers present in a given system to be cleaved. This allows for fine-grain control of protecting group removal and subsequent polymer growth.

Linkers attached to a solid support, or substrate, may be selectively activated by controlling local conditions on a portion of the substrate. Local conditions may be controlled by changing the substrate such as by selectively activating electrodes on a microelectrode array. Precise application of heat such as by use of resistors integrated into an array may be used to selectively cleave thermolabile linkers. Lithography techniques may be used to selectively expose photolabile linkers to light. Chemical cleaving agents may also be applied to selected linkers through the use of precise application techniques such as, but not limited to, chemical inkjet printing. Thus, out of a large number of linkers having the same or similar chemical structure, a selected subset may be cleaved by changing the local conditions for only those linkers.

Linkers may also be selectively cleaved by enzymes that recognize specific linker sequences. Linkers may be made out of oligonucleotides or polypeptides which are subject to sequence-specific cleavage by enzymes such as restriction endonucleases and peptidases. A set of linkers, such as linkers on a solid support structure, may include different sets of linkers each having a different sequence that is recognized by a different enzyme. Selective cleavage is achieved by exposing the linkers to the specific enzyme which will cleave only those linkers having a sequence recognized by the specific enzyme.

Selectively-cleavable linkers may be used in any number of different types of applications such as attaching molecules to a substrate or attaching a protecting group to the end of a growing polymer. When used as “surface linkers” for attaching molecules to a substrate, selectively controlling the cleavage of the linkers allows for selective release of the attached molecules. When used as “chain linkers” for attachment of a protecting group, selectively controlling the cleavage of the linkers allows for selection of specific molecules to be unprotected and modified. Use of selectively-cleavable linkers such as electrochemically-cleavable linkers, thermolabile linkers, or photolabile linkers provides controllable linkers without creating chemical waste or cross-reactions with molecules attached to the linkers.

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter nor is it intended to be used to limit the scope of the claimed subject matter. The term “techniques,” for instance, may refer to system(s) and/or method(s) as permitted by the context described above and throughout the document.

BRIEF DESCRIPTION OF THE DRAWINGS

The Detailed Description is set forth with reference to the accompanying figures. In the figures, the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The use of the same reference numbers in different figures indicates similar or identical items.

FIG. 1 illustrates a polymer connected to a substrate by a first linker and connected to a protecting group by a second linker.

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FIG. 2 illustrates an array with attached polymers belonging to separate groups.

FIG. 3 illustrates an implementation of a surface linker.

FIG. 4 illustrates an implementation of a linker that is a surface linker or a chain linker.

FIG. 5 illustrates an implementation of a chain linker.

FIG. 6 illustrates an implementation of a chain linker

FIG. 7 is a flow diagram showing an illustrative process for selectively detaching polymers from a substrate.

FIG. 8 is a flow diagram showing an illustrative process for controlling growth of a polymer with a protecting group attached by a selectively-cleavable linker

DETAILED DESCRIPTION

One application for the use of linkers is de novo synthesis of oligonucleotides. Oligonucleotides include both deoxyribose nucleic acid (DNA), ribonucleic acid (RNA), and hybrids containing mixtures of DNA and RNA. DNA includes nucleotides with one of the four natural bases cytosine (C), guanine (G), adenine (A), or thymine (T) as well as unnatural bases, noncanonical bases, and modified bases. RNA includes nucleotides with one of the four natural bases cytosine, guanine, adenine, or uracil (U) as well as

unnatural bases, noncanonical bases, and modified bases. Currently, de novo synthesis of oligonucleotides is most commonly implemented as solid-phase synthesis using the phosphoramidite method which combines phosphoramidite building blocks in the presence of a tetrazole catalyst. Nucleoside phosphoramidites are derivatives of natural or synthetic nucleosides. A phosphoramidite is a normal nucleotide with protecting groups, such as a trityl group, added to its reactive amine, hydroxyl, and phosphate groups. These protecting groups prevent unwanted side reactions and force the formation of the desired product during synthesis. Synthesis begins with a single phosphoramidite tethered to a solid support by a linker such as a succinyl linker coupled to a long chain alkylamine spacer.

The phosphoramidite method uses a cycle of four different chemistry mixtures to add each individual nucleoside in a 3' to 5' synthesis direction. First, a dimethoxytrityl (DMT)-protected nucleoside phosphoramidite that is attached to a solid support is deprotected by removal of the DMT using trichloroacetic acid (TCA) in a deblocking step. This reveals a free 5'-hydroxyl group. Second, in an activation step a new DMT-protected phosphoramidite is coupled to the 5'-hydroxyl group of the growing oligonucleotide chain to form a phosphite triester. Third, a capping step acetylates any remaining unreacted 5' hydroxyl groups, making the unreacted oligonucleotide chains inert to further nucleoside additions and preventing one source of base deletions. Capping is accomplished by adding an acetylating reagent composed of acetic anhydride and N-methyl imidazole. This reagent reacts only with free hydroxyl groups to irreversibly cap the oligonucleotides in which coupling failed. Fourth, an iodine oxidation converts the phosphite to a phosphate, producing a cyanoethyl-protected phosphate backbone stabilizing the phosphate linkage between the monomers in the growing oligonucleotide chain. The DMT protecting group is removed to allow the cycle to continue and add the next nucleoside.

The protection of the exocyclic amino groups on adenine, cytosine, and guanine has to be orthogonal to that of the 5'-hydroxy group because the DMT is removed at the end of each synthetic cycle. One technique is to use base-labile

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protecting groups on the exocyclic amino groups to create orthogonality with the TCA cleavage of the DMT protecting group.

After synthesis is complete, all the protecting groups are removed and the single-stranded oligonucleotide is cleaved from the solid support. Coupling to the solid support is achieved by covalently attaching non-nucleoside linkers to reactive amino groups on the surface of the solid support. A phosphoramidite respective to the 3'-terminal nucleoside residue is coupled to the linker in the first synthetic cycle of oligonucleotide chain assembly using the standard protocols. These non-nucleoside linkers are typically cleaved by using a mixture of bases such as gaseous ammonia, aqueous ammonium hydroxide, and aqueous methylamine.

Solid-phase synthesis of peptides (SPSS) is another application for the use of linkers. Peptide synthesis is the production of peptides, compounds where multiple amino acids are linked via amide bonds, also known as peptide bonds. Amino acids as used herein include all 20 standard amino acids, all 22 natural amino acids, non-proteinogenic amino acids, and D-isomers. Peptides are chemically synthesized by the condensation reaction of the carboxyl group of one amino acid to the amino group of another. Protecting group strategies are usually necessary to prevent undesirable side reactions with the various amino acid side chains. Chemical peptide synthesis most commonly starts at the carboxyl end of the peptide (C-terminus) and proceeds toward the amino-terminus (N-terminus).

SPSS allows the rapid assembly of a peptide chain through successive reactions of amino acid derivatives on an insoluble porous support. The general SPSS procedure is one of repeated cycles of alternate N-terminal deprotection and coupling reactions. First, an amino acid is linked to the solid support with a covalent bond between the carbonyl group of the amino acid and solid support. The bond coupling the amino acid to the solid support may be an amido or an ester bond. Subsequently, the amine is deprotected and then coupled with the free acid of the second amino acid. This cycle repeats until the desired sequence has been synthesized. SPSS cycles may also include capping steps that block the ends of unreacted amino acids from reacting. The protecting groups for the amino groups used in peptide synthesis are fluorenylmethyloxycarbonyl group (Fmoc) and tert-butyloxycarbonyl (Boc). Reactive side chains of amino acids are also protected with protecting groups that are orthogonal to the protecting group used for the amino group. At the end of the synthesis, the peptide is cleaved from the solid support while simultaneously removing all protecting groups using a strong acid such as trifluoroacetic acid or a nucleophile.

The Fmoc protecting group is base-labile. It may be removed with a dilute base such as piperidine. If Fmoc is used, the side-chain protecting groups may be provided by ester, ether, and urethane derivatives of tert-butanol. These side-chain protecting groups are removed by treatment with trifluoroacetic acid (TFA), which also cleaves the bond anchoring the peptide to the solid support. The Boc protecting group is removed with a mild acid (usually dilute TFA). The side-chain protecting groups used with Boc may be ester, ether, and urethane derivatives of benzyl alcohol. Hydrofluoric acid (HF) can be used both to deprotect the amino acid side chains and to cleave the peptide from the resin support.

This disclosure provides techniques and structures for selectively-cleavable linkers that can be used with oligonucleotide synthesis, DNA sequencing-by-synthesis, polypeptide synthesis, and other biochemical processes. Addi-

tional illustrative applications for selectively-cleavable linkers exist in fields such as DNA origami and DNA data storage.

DNA origami is the nanoscale folding of DNA to create non-arbitrary two- and three-dimensional shapes at the nanoscale. The specificity of the interactions between complementary base pairs (bp) makes DNA a useful construction material, through the design of its base sequences. Short segments of DNA called “staples” hybridize to longer strands of DNA and change the shape of the longer strands. Selectively cleavable linkers made of oligonucleotides, or other material, can be used to precisely control the separation of structures made through DNA origami. For example, individual staples may be controllably released by including a selectively cleavable linker within a staple.

DNA data storage uses pools of synthetic oligonucleotides to encode digital information from computer files. The sequence of nucleotides on strands of DNA encodes the zeros and ones representing any type of digital information. Oligonucleotide synthesis is used to create the strands of DNA for data storage. Selectively-cleavable linkers can control the release of newly-synthesized DNA strands from a solid support in a way that corresponds to the digital information encoded by the DNA strands. For example, all the DNA strands that contain digital information from the same computer file may be released from the solid support together and processed together. Other DNA strands that are attached to the same solid support may be retained and released later. There are numerous other applications for selectively cleavable controllable linkers

FIG. 1 shows an illustrative diagram 100 of substrate 102 on which a surface linker 104 and polymer chain 106 are attached. The polymer chain 106 may be an oligonucleotide, polypeptide, or other polymer. An individual monomer 108 of the polymer chain 106 is an individual unit of the polymer such as a nucleotide or amino acid. The monomer directly connected to the surface linker 104 is referred to as an anchored monomer 110. The anchored monomer 110 may have a covalent connection to the surface linker 104. In some implementations, the anchored monomer 110 may have a covalent connection directly to the substrate 102. In these implementations, the covalent bond itself is the surface linker 104.

The monomer of the polymer chain 106 that is farthest from the substrate 102 is referred to as the terminal monomer 112. If there is only one monomer attached to the surface linker 104, that single monomer is both the anchored monomer 110 and the terminal monomer 112. The terminal monomer 112 may have an attached chain linker 114 connected to a protecting group 116. The protecting group 116 may prevent elongation of the polymer chain. Thus, removal of the protecting group 116 may be part of each cycle of polymer extensions. Additionally or alternatively, the protecting group 116 may protect reactive groups on individual monomers such as amine, hydroxyl, and phosphate groups. A monomer 108 may have protecting groups 116 that both prevent elongation of the polymer and that protect reactive groups. If two types of protective groups are present on the monomers of a polymer chain 106 they may be orthogonal to each other so that removal of a protecting group 116 that prevents chain extension will not remove a separate protecting group 116 that protect reactive groups on one or more of the monomers 108.

Although the diagram 100 shows both a surface linker 104 and a chain linker 114 in implementations there may be only a surface linker 104 or only a chain linker 114. For example, a polymer chain 106 may be bound to the substrate

102 with a surface linker 104 without a protecting group 116 or a chain linker 114. Alternatively, a polymer chain 106 in solution may have a chain linker 114 and protecting group 116 attached without a covalent connection to the substrate 102.

Both the surface linker 104 and the chain linker 114 may be electrochemically-cleavable linkers, thermolabile linkers, photolabile linkers, chemically-labile linkers, or enzymatically-cleavable linkers. The surface linker 104 and the chain linker 114 may both be present and may be different types of linkers. In an implementation, this is preferable to create orthogonality between the connection to the substrate 102 and the connection to the protecting group 116. One technique for creating orthogonality between the surface linker 104 and the chain linker 114 is to have each type of linker cleaved at a different pH. For example, the surface linker 104 may cleave at a basic pH while the chain linker 114 may cleave at an acidic pH.

Electrochemically-cleavable linkers are cleaved by addition of electrons to a bond in the linker. The electrons may be generated by activating an electrode in the proximity to the bond in the linker that is to be cleaved. This may be referred to as “directly mediated cleavage” in which activation of the electrode directly causes cleavage of a bond in the linker without first changing a chemical intermediary. Electrochemically-cleavable linkers may include an ester linkage and a protected alcohol that when released can trigger an inter-molecular cyclization reaction onto the ester to form a lactone or an amide linkage and an amide that when released can trigger an intra-molecular cyclization reaction onto the amide to form a lactam. Removal of a protecting group can trigger the intra-molecular cyclization reaction. Illustrative examples of electrochemically-cleavable linkers are provided in FIGS. 3-6.

Thermolabile linkers are cleaved in the presence of heat. One structure that may be used to form thermolabile linkers is thermolytic hydroxyl protecting groups derived from 2-aminopyridine and its analogs. The resulting 2-pyridyl-substituted hydroxyl protecting groups can be efficiently cleaved with brief heat treatment at about 40-100° C. One example of thermolabile linkers is provided in U.S. Pat. No. 7,612,197. Another example of thermolabile linkers are linkers used in the CleanAmp™ phosphoramidite monomers available from Glen Research (Sterling, VA). The CleanAmp™ monomers have a (4-oxo-tetradecyl)-(N,N'-diisopropyl)-phosphoramidite linker located at the 3' hydroxyl group of a deoxyribose sugar.

Photolabile linkers are cleaved by a specific wavelength of light corresponding to the linker chemistry. There are a large number of known types of photo-cleavable bonds. Common classes of photolabile linkers include nitrobenzyl-based linkers, benzyl nitrile-based linkers, benzyl-based linkers, and carbonyl-based linkers. Amine-to-thiol cross-linkers are also photolabile and may be lengthened by attachment to a polyethylene glycol (PEG) chain. Amine-to-thiol bonds may be cleaved by ultraviolet (UV) light with a wavelength of about 365-405 nm. The list of functional groups that can be protected include, but are not limited to, phosphates, carboxylates, carbonates, carbamates, thiolates, phenolates, and alkoxides. Additionally, while the rate varies with a number of variables, including choice of solvent and pH, the photodeprotection is possible in both in solution and in linkers bound to a substrate 102.

One type of photolabile linker uses a UV photo-cleavable C3 spacer arm that includes a nitrobenzene sidechain. Cleavage occurs by irradiation with UV light (300-350 nm). Other examples of photolabile linkers are PC Biotin Phosphora-

midite with the formula 1-[2-Nitro-5-(6-(N-(4,4'-dimethoxytrityl))-biotinamidocaproamidomethyl)phenyl]-ethyl-2-cyanoethyl-(N,N-diisopropyl)-phosphoramidite, PC Amino-Modifier Phosphoramidite with the formula [(6-Trifluoroacetylamidocaproamidomethyl)-1-(2-nitrophenyl)-ethyl]-2-cyanoethyl-(N,N-diisopropyl)-phosphoramidite, PC Spacer Phosphoramidite with the formula [4-(4,4'-Dimethoxytrityloxy)butyramidomethyl]-1-(2-nitrophenyl)-ethyl-2-cyanoethyl-(N,N-diisopropyl)-phosphoramidite, and PC Linker Phosphoramidite with the formula 3-(4,4'-Dimethoxytrityl)-1-(2-nitrophenyl)-propan-1-yl-[(2-cyanoethyl)-(N,N-diisopropyl)]-phosphoramidite (available from Glen Research, Sterling, VA).

Chemically-labile linkers include acid-cleavable linkers and reducible linkers. Acid-cleavable linkers, such as hydrazine, cis-Aconity, and disulfide linkers, are designed to remain stable at neutral pH but undergo hydrolysis in an acidic environment. Acid-cleavable linkers generally cleaved at a pH of below about 5.0. Reducible linkers take advantage of the difference in reduction potential in basic environments. Succinic acid linker arms are used to attach oligonucleotides to solid supports are an example of reducible linkers and may be cleaved using ammonium hydroxide.

Another technique for inducing cleavage due to change in the local conditions may be referred to as "indirectly mediated cleavage." With indirectly mediated cleavage the change in the local conditions such as activation of an electrode, change in pH, change in temperature, exposure to light, addition of chemicals, etc. activates an auxiliary molecule which in turn causes cleavage of a bond in the linker. For example, the auxiliary molecule may be a photobase generator or photo acid generator into which electrons flow when exposed to photons of the appropriate wavelength that in turn releases an acid or base that cleaves a bond in the linker.

Many different types of auxiliary molecules, also referred to as redox mediators, are known to those of ordinary skill in the art. Examples include benzoquinone/hydroquinone that generates acid, ceric ammonium nitrate that is an electron mediator (shuttles electrons and helps coordinate with molecules), triaryl amines for oxidative cleavage of benzylethers or C—S bonds, transition metals-allylic acetate cleavage, and iodobenzene which can generate a base electrochemically. Examples of redox mediators and their use are provided in Robert Francke and R. Daniel Little, *Redox catalysis in organic electrosynthesis: basic principles and recent developments*, Chem. Soc. Rev., 2014, 43, 2492.

Enzymatically-cleavable linkers are cleaved by an enzyme that recognizes the specific sequence of the linker. Enzymatically-cleavable linkers may be formed from oligonucleotides in which case they will be cleaved by enzymes that are site-specific nucleases. Linkers formed from polypeptides are cleaved by peptidases.

There are many types of site-specific nucleases that may be used to cleave oligonucleotide sequences. Examples of the site-specific nucleases include restriction endonucleases, homing endonucleases, zinc finger nucleases, transcription activator-like effect in nucleases (TALONs), or clustered regularly interspaced short palindromic repeats (CRISPR)/Cas complexes. Each of these types of site-specific nucleases are well known to those having ordinary skill in the art.

Restriction enzymes (restriction endonucleases) are present in many species and are capable of sequence-specific binding to DNA (at a target or recognition site), and cleaving DNA at or near the site of binding. Over 3000 restriction enzymes have been studied in detail, and more than 600 of

these are available commercially. All types of restriction enzymes recognize specific short DNA sequences and carry out the endonucleolytic cleavage of DNA to give specific fragments with terminal 5'-phosphates.

Homing endonucleases (HEs), which are also known as meganucleases, are a collection of double-stranded DNases that have large, asymmetric recognition sites (12-40 bp and coding sequences that are usually embedded in either introns or inteins. Introns are spliced out of precursor RNAs, while inteins are spliced out of precursor proteins. Unlike restriction endonucleases, HEs tolerate some sequence degeneracy within their recognition sequence. Thus, single base changes do not abolish cleavage but reduce its efficiency to variable extents. As a result, their observed sequence specificity is typically in the range of 10-12 bp.

Zinc finger nucleases (ZFNs) are synthetic proteins consisting of an engineered zinc finger DNA-binding domain fused to the cleavage domain of the FokI restriction endonuclease. ZFNs consist of a DNA-binding zinc finger domain (composed of three to six fingers) covalently linked to the non-specific DNA cleavage domain of the bacterial FokI restriction endonuclease. ZFNs can bind as dimers to their target DNA sites, with each monomer using its zinc finger domain to recognize a half-site. Dimerization of ZFNs is mediated by the FokI cleavage domain which cleaves within a five or six nucleotide "spacer" sequence that separates the two inverted "half sites."

TALENs are restriction enzymes that can be engineered to cut specific sequences of DNA. They are made by fusing a TAL effector DNA-binding domain to a DNA cleavage domain (i.e., a nuclease which cuts DNA strands). Transcription activator-like effectors (TALEs) can be engineered to bind practically any desired DNA sequence, so when combined with a nuclease, DNA can be cut at specific locations.

In the CRISPR/Cas nuclease system, the CRISPR locus encodes RNA components of the system, and the Cas (CRISPR-associated) locus, encodes proteins. CRISPR loci in microbial hosts contain a combination of CRISPR-associated (Cas) genes as well as non-coding RNA elements capable of programming the specificity of the CRISPR-mediated polynucleotide cleavage.

The Type II CRISPR is one of the most well-characterized systems and carries out targeted double-stranded breaks in four sequential steps. First, two non-coding RNAs, the pre-crRNA array and tracrRNA, are transcribed from the CRISPR locus. Second, tracrRNA hybridizes to the repeat regions of the pre-crRNA and mediates the processing of pre-crRNA into mature crRNAs containing individual spacer sequences. Third, the mature crRNA:tracrRNA complex directs Cas9 to the target DNA via Watson-Crick base-pairing between the spacer on the crRNA and the protospacer on the target DNA next to the protospacer adjacent motif (PAM), an additional requirement for target recognition. In engineered CRISPR/Cas9 systems, gRNA also called single-guide RNA ("sgRNA") may replace crRNA and tracrRNA with a single RNA construct that includes the protospacer element and a linker loop sequence.

Cas9 mediates cleavage of target DNA to create a DSB within the protospacer. Activity of the CRISPR/Cas system in nature comprises three steps: (i) insertion of alien DNA sequences into the CRISPR array to prevent future attacks, in a process called 'adaptation,' (ii) expression of the relevant proteins, as well as expression and processing of the array, followed by (iii) RNA-mediated interference with the alien polynucleotide. The alien polynucleotides come from viruses attaching the bacterial cell. Thus, in the bacterial

cell, several of the so-called 'Cas' proteins are involved with the natural function of the CRISPR/Cas system and serve roles in functions such as insertion of the alien DNA, etc.

CRISPR may also function with nucleases other than Cas9. Two genes from the Cpf1 family contain a RuvC-like endonuclease domain, but they lack Cas9's second HNH endonuclease domain. Cpf1 cleaves DNA in a staggered pattern and requires only one RNA rather than the two (tracrRNA and crRNA) needed by Cas9 for cleavage. Cpf1's preferred PAM is 5'-TTN, differing from that of Cas9 (3'-NGG) in both genomic location and GC-content. Mature crRNAs for Cpf1-mediated cleavage are 42-44 nucleotides in length, about the same size as Cas9's, but with the direct repeat preceding the spacer rather than following it. The Cpf1 crRNA is also much simpler in structure than Cas9's; only a short stem-loop structure in the direct repeat region is necessary for cleavage of a target. Cpf1 also does not require an additional tracrRNA. Whereas Cas9 generates blunt ends 3 bp upstream of the PAM site, Cpf1 cleaves in a staggered fashion, creating a five nucleotide 5' overhang 18-23 bp away from the PAM.

Other CRISPR-associated proteins besides Cas9 may be used instead of Cas9. For example, CRISPR-associated protein 1 (Cas1) is one of the two universally conserved proteins found in the CRISPR prokaryotic immune defense system. Cas1 is a metal-dependent DNA-specific endonuclease that produces double-stranded DNA fragments. Cas1 forms a stable complex with the other universally conserved CRISPR-associated protein, Cas2, which is part of spacer acquisition for CRISPR systems.

There are also CRISPR/Cas9 variants that do not use a PAM sequence such as NgAgo. NgAgo functions with a 24-nucleotide ssDNA guide and is believed to cut 8-11 nucleotides from the start of this sequence. The ssDNA is loaded as the protein folds and cannot be swapped to a different guide unless the temperature is increased to non-physiological 55° C. A few nucleotides in the target DNA are removed near the cut site.

Peptidases or proteases are catalytically active proteins (enzymes) that cleave peptide bonds in proteins and peptides by hydrolysis. There are six catalytic types (serine, cysteine, threonine, aspartic, glutamic, and metallo). Peptidases can be classified by sequence similarities into approximately 250 families, and these can be arranged into approximately 60 clans by comparing tertiary structures. In an implementation, the protease or peptidase is an exopeptidase or an endopeptidase. In an implementation, the protease or peptidase is an endopeptidase or endopeptidase. The protease or peptidase may include trypsin, chymotrypsin, pepsin, papain any cathepsin (e.g., cathepsin B, L, D, K, or G). A skilled artisan will recognize that methods for the enzymatic digestion of polypeptides are well known in the art.

The enzymatically-cleavable linkers may be attached to the surface of the substrate **102** during a surface coating or functionalization step. Additionally or alternatively, the oligonucleotides or polypeptides that form an enzymatically-cleavable linker may be synthesized in situ directly onto the surface of the substrate **102**. Oligonucleotide linkers are generally single-stranded DNA. However, most site-specific nucleases recognize double-stranded DNA. Therefore, cleavage of an oligonucleotide linker may include contacting the linker with a short primer sequence that hybridizes and forms a double-stranded oligonucleotide. This double-stranded region is then recognized by a site-specific nuclease. Upon cleavage of a recognition site, the oligonucleotide linker and the primer are released.

For example, a surface linker **104** that is a DNA strand with the nucleotide base sequence GAATTC will be contacted with a primer that has the reverse-complementary nucleotide base sequence of CTTAAG. In an implementation, this will be performed by flowing an excess of primers across the surface of the substrate **102** so that hybridization occurs for all surface linkers **104** having the complementary nucleotide base sequence. In this example, a restriction enzyme will cleave the double-stranded DNA (dsDNA) between the guanine and adenine bases. This would then release the polymer chain **106** with the sequence AATTC attached to the anchored monomer **110**. The complementary portion of the primer, the TTAAG sequence, may remain hybridized to the portion of the surface linker **104** that is attached to the anchored monomer **110** or it may disassociate depending on changes in conditions such as temperature and salt concentration.

If an enzymatically-cleavable surface linker **104** is formed from the same monomers as the polymer chain **106**, there is a possibility that the enzyme which cleaves the surface linker **104** may also cleave a portion of the polymer chain **106**. To avoid this, the recognition sequence in the surface linker **104** may be selected to be different from any sequences found in the polymer chain **106**. Alternatively, the sequences of the polymer chains **106** may be designed to not include any sequences that are recognized by enzymes used for cutting any of the surface linkers **104** attached to the substrate **102**.

For oligonucleotide surface linkers **104** attached to oligonucleotides, existing design techniques and software for avoiding primer collisions may be adapted to identify oligonucleotide sequences for the surface linkers **104** that are not found in any of the attached polymer chains **106**. If the surface linker **104** and the polymer chain **106** are polypeptides, then the surface linker **104** may be designed to include a sequence that is not present in any of the polypeptide chains using techniques similar to that for avoiding DNA primer collision.

The surface linker **104** may also be created with unnatural or modified bases to create a unique recognition site that will not be present in any of the attached polymer chains **106**. For example, oligonucleotides forming the surface linker **104** may be methylated so that a restriction enzyme that only acts on methylated DNA will not cleave portions of the polymer chain **106** even if the sequence of nucleotide bases is the same as the surface linker **104**. Type IIM restriction endonucleases, such as DpnI, recognize and cut only methylated DNA. The recognition site for DpnI is G_mA/TC which is also referred to as the dam sequence since it is recognized by dam methylase. DpnI does not cleave unmethylated DNA and will cleave hemimethylated DNA (with only one adenine methylated) 60-fold more slowly than fully adenomethylated recognition sequences.

Alternatively, the polymer chain **106** may be created with unnatural or modified bases such as methylated bases to prevent cleavage of oligonucleotides in the polymer chain **106** by a site-specific nuclease that cleaves the surface linker **104**. Methylation may be applied selectively to the polymer chain **106** to those regions of DNA that have the same nucleotide base sequence as the surface linker **104**. A methyltransferase may be used to methylated all or portions of the polymer chain **106** after it has been synthesized or methylated bases may be incorporated into the polymer chain **106** as the oligonucleotide is growing.

The substrate **102** may be a two-dimensional substrate such as an array or a three-dimensional substrate such as a bead or microsphere. Arrays may be formed from silicon

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chips, glass, an insoluble polymer, or other material. A glass substrate **102** may be formed from controlled porous glass (CPG) with pore sizes between about 50 and 300 nm. A silica substrate has a relatively homogeneous chemical surface echoing modified using well-developed silanization chemistry. Beads may be formed from silicon, glass, polystyrene, polymeric resins, latex, etc. Magnetic beads may also be used in which magnetic material such as ferrite crystals are included in suspension of styrene/divinylbenzene monomers. The magnetic properties provide additional handling and manipulation options for magnetic beads.

The substrate **102** may be modified or functionalized prior to attachment of the surface linker **104**. For example, the substrate **102** may be silanized by covering the surface of the substrate **102** with organofunctional alkoxy silane molecules. This treatment creates a uniform layer of primary amines or epoxides that are available to react with the surface linker **104**. Mineral components like glass and metal oxide surfaces can all be silanized because they contain hydroxyl groups that attack and displace the alkoxy groups on the silane thus forming a covalent —Si—O—Si— bond. Modifications of the surface of the substrate **102** that provide a location for covalent coupling include the addition of carboxylic acid groups, primary aliphatic amines, aromatic amines, chloromethyl (vinyl benzyl chloride), secondary amine groups, hydrazide groups, aldehydes, hydroxyl groups, thiol groups, epoxy groups, mercaptosilanes, etc.

In an implementation, the substrate **102** may include multiple spatially-addressable electrodes. Each electrode may be independently addressable allowing the creation of arbitrary and variable voltage microenvironments across the surface of the array. This type of substrate **102** may be referred to as a “microelectrode array.” The microelectrode density may be approximately 1000 microelectrodes/cm², approximately 10,000 microelectrodes/cm², or a different density. One example of a microelectrode array is provided in Bo Bi et al., *Building Addressable Libraries: The Use of “Safety-Catch” Linkers on Microelectrode Arrays*, 132 J. Am. Chem. Soc. 17, 405 (2010).

Oxidation of an electrolyte solution after the application of current to microelectrodes liberates acid at the anodes; concomitant reduction at the cathodes consumes acid. In an implementation, the cathode is a palladium cathode. The ions and radicals generated by these redox reactions at the electrode surfaces move away from the electrodes and to the substrate through a combination of diffusion, migration and convection effects. During this transit time the electrode products may further react. Once the primary or secondary products reach the substrate, they may either react to remove the protecting group **116** by cleaving a chain linker **114** or liberating the polymer chain **116** from the substrate **102** by cleaving the surface linker **104**. One illustrative type of bond that can be cleaved by a redox reaction is an ester that forms a carbonyl group.

The electrodes in a microelectrode array may be implemented using complementary metal-oxide-semiconductor (CMOS) integrated circuits. CMOS circuits use a combination of p-type and n-type metal-oxide-semiconductor field-effect transistor (MOSFETs) to implement logic gates and other digital circuits. Although CMOS logic can be implemented with discrete devices for demonstrations, commercial CMOS products are integrated circuits composed of up to billions of transistors of both types, on a rectangular piece of silicon of between 10 and 400 mm². A series of controllable gates/transistors implemented with CMOS circuits can be controlled to inject charge at any location on the substrate **102**.

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Protecting group **116** may be any type of protecting group used in organic synthesis. See, e.g., Greene et al., *Protective Groups in Organic Synthesis*, 3rd Ed., John Wiley & Sons, Inc., New York, N.Y. (1999). Protecting groups are often used to prevent a particular functional group or part of a molecule (e.g., a phosphite, a phosphate, an amine, a carboxylic acid, a thiol, a hydroxyl, a heterocycle, etc.) from reacting under certain reaction conditions (e.g., a chemical reaction in which an unprotected part of the same molecule undergoes a synthetic transformation). Hydroxyl-protecting groups are among the most commonly used protecting groups in organic synthesis. It will be appreciated that the protecting group **116** should be chosen based on the type of substituent that is protected, the structure of the molecule for which the protecting group **116** is used, the reaction conditions used, the type of solvent used, the conditions required for removing the protecting group, and the like. One of skill in the art may choose from among different protecting groups to protect functional groups. Reaction conditions that influence the choice of protecting group can typically include the pH, the temperature, the relative reactivities of the reactants and/or products, and the like.

Protecting groups for hydroxyls can include, for example, silyl ethers (e.g., trimethyl silyl, triethyl silyl, tert-butyldimethyl silyl, tert-butyldiphenylsilyl, dimethylphenylsilyl, and diphenylmethylsilyl), benzyl carbonates, trityl, monomethoxytrityl, dimethoxytrityl, esters (e.g., acetate, benzoate, and the like), pixyl, tert-butyloxycarbonyl, a tetrahydropyranyl group, photolabile protecting groups and the like. When the hydroxyl is a sugar hydroxyl, suitable protecting groups can include, for example, pixyl, acetyl, t-butyldimethylsilyl (CBDMS), trityl, monomethoxytrityl (“MMT” or “MMTr”), dimethoxytrityl (“DMT” or “DMTr”), and the like. Protecting groups for nitrogen include, for example, amides (e.g., trifluoroacetyl, acetyl, phenoxyacetyl, benzoyl, and isobutyryl), carbamates (e.g., tert-butyloxycarbonyl, (4-nitrophenyl)ethyloxycarbonyl, and N-benzyloxycarbonyl), trityl, amidines and the like.

Two protecting groups used in organic synthesis are Fmoc and Boc. Fmoc is a base-labile protecting group used in organic synthesis. Fmoc carbamate is frequently used as a protecting group for amines, where the Fmoc group can be introduced by reacting the amine with fluorenylmethyloxycarbonyl chloride (Fmoc-Cl). Another method for introducing the Fmoc group is through 9-fluorenylmethylsuccinimidyl carbonate (Fmoc-OSu), which may itself be obtained by the reaction of Fmoc-Cl with the dicyclohexylammonium salt of N-hydroxysuccinimide. Fmoc may be cleaved by bases typically in a solution of piperidine or in tetra-n-butylammonium fluoride in the presence of dimethylformamide.

Boc can be used in the protection of amines and amino acids under either aqueous or anhydrous conditions, by reaction with a base and the anhydride Boc₂O. Active esters and other derivatives such as Boc-ONH₂ and Boc-N₃ can also be used. The Boc group is stable towards most nucleophiles and bases. tert-Butyl carbamates are cleaved under anhydrous acidic conditions with the production of tert-butyl cations. Scavengers such as thiophenol may prevent nucleophilic substrates from being alkylated. Removal of the BOC in amino acids can be accomplished with strong acids such as TFA (neat or in dichloromethane) or with HCl in methanol.

The protecting group **116** may also protect the terminal monomer **112** from addition of subsequent monomers **108** by steric hindrance. Thus, the steric bulk of the protecting group **116** rather than its chemistry may prevent polymer-

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ization reactions such as the formation of phosphodiester bonds in DNA synthesis or peptide bonds in polypeptide synthesis. Steric hindrance results from repulsive forces between overlapping electron clouds.

FIG. 2 shows an array **200** partially covered with attached polymer chains **202**. The attached polymer chains **202** are each attached to the array **200** by a surface linker **204**. Upon cleavage of the surface linker **204**, the respective polymer from the attached polymer chain **202** is released and becomes a free polymer **206**. The polymer chain **202** and the surface linker **204** may be the same or similar to the polymer chain **106** and the surface linker **101** introduced in FIG. 1. Depending on the size of the array **200** and the density with which the attached polymer chain **202** are present, the array **200** may be covered with millions or billions of individual surface linkers **204** and attached polymer chains **202**.

The attached polymer chain **202** may be associated with one or more different groups. The groupings may be spatial or logical based on the content of the attached polymers. Spatial groupings are based on the location of placement on the surface of the array **200**. Spatial groupings may be contiguous or discontinuous and take any possible shape. For example, spatial groupings may be in a checkerboard pattern, a square, a rectangle, a circle, etc. Logical groupings are based on a common feature possessed by the polymers that are grouped together. For example, if the attached polymer chains **202** are used to store digital information, the logical groupings may be based on the content of the information stored. In an implementation, all of the attached polymer chains **202** storing a portion of the digital information from the same computer file may be grouped together. Logical groupings may, but need not necessarily, share a spatial grouping. For example, all of the attached polymer chains **202** that belonged to a given logical grouping may be placed together in a contiguous spatial grouping such as a rectangle-shaped grouping. However, logical groupings need not have any defined spatial grouping. Logical groupings may be distributed randomly or pseudo-randomly across the surface of the array **200**.

In this illustration, the attached polymer chains **202** are shown as being divided into four different groups: group A **208**, group B **210**, group C **212**, and group D **214**. However, in an actual implementation the number of groups may be fewer or greater such as many thousands or hundreds of thousands of separate groups. Group D **214** illustrates selective cleavage of the surface linkers **204** to release the polymers in group D **214** without releasing the polymers from any of the other groups. The surface linkers **204** for each group may be cleaved separately from the surface linkers **204** of the other groups. Thus, the surface linkers **204** for any of group A **208**, group B **210**, group C **212**, or group D **214** may be separately and independently cleaved. Group A **208**, group B **210**, group C **212**, and group D **214** may represent either spatial or logical groupings. Spatial and logical groupings may also be combined on the same array **200**. For example, group B **210** may represent a spatial grouping while group C **212** may represent a logical grouping.

Each grouping of attached polymer chains **202** may be, but is not necessarily, contiguous. For example, group C **212** is split across two separate regions. Any group may be split into any number of regions limited only by the resolution with which the surface linkers **204** are able to be selectively cleaved.

Cleavage of the surface linkers **204** may be by any of the mechanisms discussed above such as electrochemical cleavage, thermal cleavage, photocleavage, chemical cleavage, or

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enzymatic cleavage. If the array **200** is implemented as a microelectrode array, activation of specific sets of spatially-addressable electrodes on the microelectrode array can implement selective release of one or more groups of attached polymer chains **202**. In this implementation, all of the surface linkers **204** may have the same structure and the selective activation is controlled by the microelectrodes within the array **200**. The array **200**, when implemented as a microelectrode array, is coupled to a power source for powering the electrodes and to control circuitry for directing the current to specified ones of the spatially-addressable electrodes. In an implementation, array **200** may be implemented with CMOS circuitry.

A microelectrode array may also implement selective cleavage of thermally-labile linkers. Electrodes in the microelectrode array may be coupled to localized heat sources such as spatially-addressable resistors. Heat created by the resistor can cleave thermally-labile linkers within sufficient proximity to the resistor that the temperature of the linker rises above its cleavage temperature. The resolution of the array **200** may be determined by the thermal properties of the array **200** itself and of any solution covering the surface of the array **200**. Thus, thermal diffusion may cause surface linkers **204** that are not directly above a given spatially-addressable resistor to also cleave.

Thermally-labile linkers may be cleaved by heat sources other than resistors. For example, lasers or focused ultrasound beams may be used to heat specific portions of the array **200**.

Photolabile linkers attached to the array **200** may be selectively cleaved by controlling exposure to a light source. For example, UV light directed through a photolithographic mask will irradiate and cleave only those surface linkers **204** that are exposed to the UV light. Thus, the shape of a photolithographic mask can control which group of attached polymer chains **202** are released. With photolabile linkers, all of the surface linkers **204** may have the same configuration and the control of cleavage is implemented by controlling exposure to light. In addition to use of a photolithographic mask, exposure to light may also be controlled by using focused light sources such as a laser or array of light emitting diodes (LEDs) that directs light only on to specified portions of the array **200**.

Chemically-labile linkers generally cleaved across an entire array **200** when the array is exposed to the appropriate chemical for cleaving the surface linkers **204**. However, selective application of the chemical cleaving agent to some portions of the array **200** may be used to implement selective control of chemically-labile linkers. Instead of applying a chemical cleaving agent across an entire array **200** or collection of beads, the chemical cleaving agent may be applied to only selected areas by a spatially-controlled application technique. The chemical cleaving agent may be applied to the array **200** by location-specific printing such as inkjet-like printing. Chemical inkjet printing uses techniques similar to conventional printing to place nanoliter volumes of reagents at specified locations on a two-dimensional surface. The appropriate acid, base, or other chemical cleaving agent may be applied to specific surface linkers **204** by use of inkjet printing. This provides site-specific cleavage of surface linkers **204** and release of attached polymer chains **202** at the location where the reagent is dispensed.

Any type of chemical inkjet printing may be adapted for use with this disclosure. Inkjet printing can be divided into two categories: (1) drop-on-demand (DoD) or impulse inkjet, where droplets are generated when required; and (2) continuous inkjet, in which droplets are deflected from a

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continuous stream to a substrate when needed. Inkjet printing can be further subdivided according to the specific means of generating droplets, such as piezoelectric, thermal and electrostatic. Droplet size involves, typically, volumes ranging from 1.5 pL to 5 nL at a rate of 0-25 kHz for drop-on-demand printers (and up to 1 MHz for continuous printheads).

Electrohydrodynamic jet printing (EHJP) is another printing technology that may be used. EHJP is a high-resolution printing technology where the printed liquid is driven by an electric field. Exposure to an electric field causes mobile ions in a polarizable liquid to accumulate at the liquid surface. Deposited droplets can be as small as 240 nm with spatial accuracy in the hundreds of nm, which is typically an order of magnitude smaller than other inkjet printing technologies. Such small droplet sizes dispense less material with more spatial control, which allows for more selectivity in controlling the cleavage of specific surface linkers **204**.

Enzymatically-cleavable linkers have cleavage specificity built into the sequence of the linkers (i.e., the oligonucleotide or polypeptide chain that is the linker). Thus, controlling the cleavage of enzymatically-cleavable linkers is done not by changing the local environment around the surface linkers **204**, but by contacting the surface linkers **204** with the appropriate enzyme. Thus, during design construction of the array **200**, all polymers that belong to the same group are attached to the array **200** by surface linkers **204** that have the same sequence or a similar sequence that is cleaved by the same enzyme. Different groups such as group A **208**, group B **210**, group C **212**, and group D **214** may be differentiated by use of surface linkers **204** that have sequences which are recognized by different enzymes. Those of ordinary skill in the art will appreciate that design of the enzymatically-cleavable linkers for different groups on the same array **200** will consider cross-reactivity and select linker sequences that are not cleaved by any of the enzymes used for cleaving linkers of the other groups.

Preventing cross-reactivity between surface linkers **204** used with different groups may also be achieved by using different types of enzymatically-cleavable linkers. For example, a first group of polymers could be attached to the array **200** by oligonucleotides that are cleaved by a restriction endonuclease and a second group of polymers could be attached to the array **200** by polypeptides that are cleaved by a peptidase. The use of different types of polymers for the enzymatically-cleavable linkers (i.e., oligonucleotides and polypeptides) provides orthogonality between these two sets of surface linkers **204**.

FIG. 3 shows an illustrative surface linker **300** that may be attached to the substrate **102**. R_1 and R_2 are independently selected amino acid side chains that include one or more amino acids. This polypeptide chain may be synthesized using conventional solid-state peptide synthesis techniques. R_3 is hydrogen (H) or an allyl group ($H_2C=CH-CH_2-$). This surface linker **300** uses a two-stage cleavage process. Two-stage cleavable linkers are not limited to the specific structure shown in FIG. 3 but include any linker structure that has a first cleavable group which protects a second cleavable group from cleavage. The first cleavable group may be attached by a first cleavage location **302** which in an implementation is an electrochemically cleavable bond that may be cleaved by oxidation caused by activating a spatially-addressable electrode on a microelectrode array. Cleavage of the second cleavable group cleaves the surface linker **300** and releases any bound polymer from the substrate **102**. The second cleavable group may be any group that reacts to a different cleavage trigger than the first

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cleavable group. In an implementation, the second cleavage location **304** is a peptide bond that is cleaved by peptidase which recognizes the sequence of polypeptides extending from R_1 through R_2 . A peptide bond is an amide type of covalent chemical bond linking two consecutive alpha-amino acids from C1 (carbon number one) of one alpha-amino acid and N_2 (nitrogen number two) of another along a polypeptide or protein chain.

The surface linker **300** may also include a spacer **306**. The spacer **306** may be an organic spacer (e.g., an aliphatic spacer, an alkyl spacer, an aromatic spacer, an alkylene glycol, a polyethylene glycol, a carbohydrate (e.g., a sugar)), or the like. In an implementation, the spacer **306** may be a PEG spacer. PEG spacers are well-known inert spacers used in many biotechnological applications. If the spacer **306** is omitted, a functional group on the polymer chain **106** may connect directly to the nitrogen on the surface linker **300**.

Removal of the allyl group by cleavage at the first cleavage location **302** exposes the peptide bond to the peptidase. Electrochemical control of the first cleavage location **302** allows for site-specific cleavage of the surface linker **300**. When contacted by the peptidase after cleavage at the first cleavage location **302**, the peptidase cleaves at the second cleavage location **304** and releases the polymer chain **106**. However, the peptidase only cleaves those surface linkers **300** for which the protecting allyl group was removed. Thus, even if the peptidase is applied uniformly across the surface of an array covered with surface linkers that have identical structures, only those surface linkers **300** for which cleavage has occurred at the first cleavage location **302** will be cleaved by the peptidase.

Cleavage of the polypeptide chain at the second cleavage location **304** leaves a scar attached to the polymer chain **106**. The scar includes the spacer **306** and the peptide of polypeptides that make up R_2 . If necessary, the scar may be removed through subsequent processing of the polymer chain **106**.

In an implementation, an array such as the array **200** introducing FIG. 2 may be covered with the surface linker **300**. Cleavage of individual ones of the surface linker **300** may be selectively controlled by activating specific electrodes in the array **200**.

FIG. 4 shows a dual-purpose linker **400** that in implementations may function as either a surface linker or a chain linker. The dual-purpose linker **400** may include an alkoxy bond to the substrate **102** if functioning as a surface linker. When functioning as a surface linker, the dual-purpose linker **400** may omit the side chain that contains the protecting group **116**. The cleavage location **402** is in an ester that is connected to the polymer chain **106** by a variable length hydrocarbon chain (m represents the number of ethylene units in the hydrocarbon chain).

If functioning as a chain linker, the dual-purpose linker **400** will include the protecting group **116**. The protecting group **116** may be any of the types of protecting groups discussed above. The protecting group **116** is connected to the remainder of the dual-purpose linker **400** by a variable group X that is either oxygen (O) or a secondary amine ($N-H$). When functioning as a chain linker, the dual-purpose linker **400** will lack a covalent connection to the substrate **102**. Thus, instead of an alkoxy bond to the substrate **102**, the dual-purpose linker **400** may end with a hydroxyl group. Alternatively, the sidechain between the ester and the substrate **102** may be truncated or omitted.

The cleavage location **402** may be cleaved by exposure to electrons created by an electrode such as an electrode on a microelectrode array (e.g., microelectrode array **200**).

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FIG. 5 shows a chain linker **500** used for adding a protecting group **116** onto the growing strand of DNA. The chain linker **500** includes a deoxy nucleotide triphosphate. Chain linker **500** is illustrated with the pyrimidine base cytosine; however, the other pyrimidine bases uracil or thiamine may be substituted. The protecting group **116** may be any of the protecting groups described previously. In one implementation, the protecting group **116** may be benzotriazole. Removal of the protecting group **116** triggers a cyclization reaction and cleavage of the chain linker **500** at cleavage location **502**.

The protecting group **116** is connected to the pyrimidine base by a variable length sidechain where m represents one or more carbon atoms (C). R₁, R₂, and R₃ may be the same or different and may be a saturated alkyl containing from about one to about 10 carbon atoms. The term "saturated alkyl" means a straight- or branched-chain saturated alkyl. Examples of saturated alkyls include methyl, ethyl, propyl, isopropyl, n-butyl, sec-butyl, octadecyl, and the like. One of R₁, R₂, and R₃ includes a DNA polymerase optionally attached via a spacer such as any of the type of spacers described in conjunction with spacer **306** of FIG. 3. Thus, in this implementation the enzyme that catalyzes growth of the oligonucleotide chain is covalently attached to a nucleotide monomer.

The DNA polymerase may be any type of DNA polymerase that can extend the growing oligonucleotide chain. In an implementation, the DNA polymerase attached at one of R₁, R₂, or R₃ may be terminal deoxynucleotidyl transferase (TdT), also known as DNA nucleotidylexotransferase (DNTE) or terminal transferase. TdT catalyzes the addition of nucleotides to the 3' terminus of a DNA molecule. Unlike most DNA polymerases, it does not require a template. The preferred substrate of this enzyme is a 3'-overhang, but it can also add nucleotides to blunt or recessed 3' ends. Cobalt is a cofactor, however, TdT catalyzes polymerization upon Mg and Mn administration in vitro.

The cleavage location **502** is an ester that is cleaved by removal of the protecting group. Activation of an electrode may cause removal of the protecting group and cleavage at the cleavage location **502**. Although this cleavage location may be several oligonucleotides away from the surface of a microelectrode array it can be affected by a microelectrode in the array. Upon electrochemical activation, removal of the protecting group and cleavage at the cleavage location **502** induces a cyclization that produces a five- or six-member heterocycle including positions numbered 1 to 5. A heterocyclic compound or ring structure is a cyclic compound that has atoms of at least two different elements as members of its ring(s). This leaves a scar attached to the pyrimidine base that includes a variable length hydrocarbon chain ending with a hydroxyl group.

FIG. 6 shows a chain linker **600** with a different structure but similar functionality to the chain linker **500** shown in FIG. 5. R₁, R₂, and R₃ may be the same or different and may be a saturated alkyl contain from about one to about 10 carbon atoms and one of R₁, R₂, and R₃ includes a DNA polymerase option attached via a spacer. The cleavage location **602** is an ester that is attached to the pyrimidine base by a variable length hydrocarbon chain. Cytosine is shown as the pyrimidine base; however, any other pyrimidine base may be substituted. Cleavage of chain linker **600** at cleavage location **602** leaves a similar scar as described above for chain linker **500**. The leaving group following cleavage at cleavage location **602** is a two-membered ring.

The scars left following cleavage of any of the surface linker **300**, the dual-purpose linker **400**, the chain linker **500**,

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the chain linker **600**, or any other linker contained in the disclosure may be removed from the polymer chain by subsequent processing. Scars attached to ssDNA strands may be removed, or more precisely diluted, by polymerase chain reaction (PCR) amplification. Copies of the ssDNA strand created by PCR amplification would be formed from nucleosides that do not have scars. Thus, none of the population of double-stranded DNA molecules created by PCR would have scars except for the molecule that incorporates the original ssDNA strand.

Illustrative Processes

For ease of understanding, the processes discussed in this disclosure are delineated as separate operations represented as independent blocks. However, these separately delineated operations should not be construed as necessarily order dependent in their performance. The order in which the processes are described is not intended to be construed as a limitation, and unless otherwise contradicted by context any number of the described process blocks may be combined in any order to implement the process or an alternate process. Moreover, it is also possible that one or more of the provided operations is modified or omitted.

FIG. 7 shows process **700** for selectively detaching polymers from a substrate by the use of selectively-cleavable linkers. Process **700** may be implemented, for example, using one of the linkers shown in FIG. 3 or FIG. 4.

At **702**, a substrate is prepared with selectively-cleavable linkers. The substrate may be the same or similar to the substrate **102** introduced in FIG. 1. The selectively-cleavable linkers may be the same or similar to the surface linker **104** introduced in FIG. 1, the surface linker **204** introduced in FIG. 2, the surface linker **300** introduced in FIG. 3, or the dual-purpose linker **400** introduced FIG. 4.

The selectively-cleavable linkers may be electrochemically-cleavable linkers, photolabile linkers, thermolabile linkers, acid-labile linkers, base-labile linkers, or enzymatically-cleavable linkers. The substrate may be any of the types of substrates discussed above such as a two-dimensional array or a three-dimensional bead. Preparation of the substrate may include functionalization of the surface such as by silanization or another technique to add reactive groups such as esters or amides to the surface of the substrate. In an implementation, the surface of the substrate may be coated with agarose or another polysaccharide.

The selectively-cleavable linkers are attached to the substrate by a reaction that forms a covalent bond between the linkers and the substrate itself or coating on the substrate. In an implementation, linkers may be attached to the substrate by an ester or amide linkage. Techniques for attaching linkers to substrates, such as ester chemistry for attaching the linker to a surface, are well-known to those having ordinary skill in the art. Known techniques for solid phase synthesis of oligonucleotides or polypeptides may be adapted for attachment of any of the linkers described in this disclosure.

The selectively-cleavable linkers attached to the substrate may all be the same (i.e., a large number of beads all coated with the same linkers) or a single substrate may be coated with two or more different types of linkers. For example, an array may be coated with a first enzymatically-cleavable linker with a recognition site for a first enzyme on a first portion of the array and a second enzymatic-cleavable linker with a recognition site for a second enzyme on a second portion of the array.

At **704**, polymers are grown on the selectively-cleavable linkers attached to the substrate. The polymers may be the same or similar to the polymer chain **106** introduced in FIG. 1. For example, the polymers may be oligonucleotides grown by conventional solid-phase oligonucleotide synthesis techniques or polypeptides grown by conventional solid-phase peptide synthesis techniques. Techniques for growing oligonucleotides include synthesis that proceeds in the 5' to 3' direction as well as in the 3' to 5' direction.

At **706**, a portion of the linkers are selectively cleaved. Selective cleaving of the linkers cleaves some but not all of the linkers on the substrate. The linkers to be selectively cleaved, and thus the polymers that will be released may be selected after synthesis by changing local conditions on a portion of the substrate.

The local conditions may be changed by activating one or more spatially-addressable electrodes on the substrate. Activating the electrodes may oxidize the linkers resulting in cleavage of covalent bonds separating the attached polymer from the substrate. Activating spatially-addressable resistors creates heat at the locations of the activated resistors. The heat can cleave thermolabile linkers in the proximity of the activated resistors. Controlled application of light to the substrate such as with optoelectronics or by applying light through a lithographic photomask can change local conditions by directing the light onto only selected portions of the substrate. The wavelength of the light may be selected to be a wavelength that cleaves the photolabile linkers.

Chemically-labile linkers may also be cleaved by selectively applying appropriate chemical cleaving agents to only portions of the substrate. For example, acid-labile linkers may be cleaved by selectively applying acid to only those portions of the substrate for which cleavage is desired. Similarly, base may be selectively applied to cleave base-labile linkers. Techniques for control application of limited volumes of chemical cleaving agents include chemical inkjet printing to place small volumes of the cleaving agents at specific positions on the surface of a two-dimensional array. Thus, the acid or the base may be selectively applied to portions of the substrate without applying acid or base to other portions of the substrate. These are examples of directly mediated cleavage.

Another technique for changing local conditions to induce cleavage of acid-labile linkers or base-labile linkers is to selectively apply light to activate a photoacid generator or a photobase generator. This is an example of indirectly mediated cleavage. Light may be applied to a portion of the substrate using any of the techniques suitable for photolabile linkers. The selective application of light limits the portions of the substrate in which acid is generated by a photoacid generator or in which base is generated by a photobase generator. Accordingly, the chemically-labile linkers are cleaved only on the portions of the substrate that receive light. Examples of photobase generators include carbamates, O-acyloximes, and ammonium salts. See e.g., Kanji Sugiyama and Masamitsu Shirai, *Photobase generators: Recent progress in application trend in polymer systems*, 34(2) Progress in Polymer Sci. 194 (2009). Examples of photoacid generators include designs based on ionic, benzyl ester, imino ester, spiropyran, and terarylene frameworks. See e.g., Collin Martin et al., *Recent progress in development of photoacid generators*, 34 J. of Biochemistry and Photobiology C: Photochemistry Reviews 41 (2018).

Selective cleavage of a portion of the linkers may be performed to release only those polymers that share a logical relationship. For example, if the polymers are oligonucleotides including digital information, selective cleavage of

the linkers may be used to release only those polymers that include digital information from the same computer file. The substrate could then be washed by removing only the oligonucleotides that include digital information for the specific computer file. These oligonucleotides may be stored together such as in a container such as an Eppendorf tube or dried onto a substrate such as filter paper. Thus, oligonucleotides that encode digital information from the same computer file are able to be stored together without contamination from oligonucleotides containing digital information for other computer files. This process could be repeated to isolate and separately store multiple groups oligonucleotides each group encoding digital information from a different computer file.

Selective cleavage of enzymatically-cleavable linkers is performed by selection of the enzyme used to cleave the linkers. If a substrate is covered with two or more sets of selectively-cleavable linkers that are cleaved by different enzymes (e.g., two sets of oligonucleotide linkers that have different recognition sequences or a set of oligonucleotide linkers cleaved by a restriction endonuclease and a set of polypeptide linkers cleaved by a peptidase) then selecting an enzyme based on the recognition sequence of the linkers to be cleaved will result in cleavage of only the selected subset of linkers. Thus, the grouping of linkers and the resolution with which different polypeptides can be separated from the substrate is based on the choice of linkers used at **702** when the substrate is prepared.

FIG. 8 shows process **800** for controlling the growth of a polymer using a protecting group that is attached with a selectively-cleavable linker. Process **800** may be implemented, for example, using one of the linkers shown in FIG. 4, FIG. 5, or FIG. 6.

At **802**, a monomer is attached to a substrate. The monomer may be the same or similar to the anchoring monomer **110** introduced in FIG. 1. The substrate may be the same or similar to the substrate **102** also introduced in FIG. 1. In some implementations, the monomer may be a nucleotide or an amino acid anchored to the substrate using conventional solid phase oligonucleotide synthesis or solid phase peptide synthesis techniques. Alternatively, the monomer may be anchored to the substrate using a selectively-cleavable linker as described in this disclosure.

At **804**, a linker to protecting group is selectively cleaved. The protecting group prevents extension of the polymer chain. The protecting group may be any of the types of protecting groups described elsewhere in this disclosure. The linker may be the same or similar to the chain linker **114** and the protecting group may be the same or similar to the protecting group **116** introduced in FIG. 1. In implementations, the protecting group may be any protecting group used with conventional oligonucleotide synthesis (e.g., 5'-DMT (4,4'-dimethoxytrityl) or peptide synthesis techniques (e.g., Fmoc or Boc).

Cleavage of the linker and removal of the protecting group exposes the monomer attached to the substrate so that an additional monomer may be added to extend a growing polymer chain. Oligonucleotide synthesis may begin with either a template ssDNA or with a linker. Synthesis can proceed in the 5' to 3' direction or, depending on specific synthesis technique, in the 3' to 5' direction. Thus, the linker may be attached to either the 5' or the 3' hydroxyl groups of a dNTP. Natural dNTPs containing the protecting group are incorporated into the growing chain using transferase enzymes such as TdT. The protecting group itself may be covalently attached to the enzyme that polymerizes the monomers such as in implementations in which the TdT

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enzyme is attached to dNTPs by a linker. The protecting group on the dNTP prevents multiple incorporation of that same base onto the growing polymer chain. Removal of the protecting group may be referred to as a “deblock step.”

At **806**, the monomer may be contacted with one or more types of activators. For example, in phosphoramidite synthesis of oligonucleotides, the diisopropylamino group of the incoming phosphoramidite monomer in the solvent acetonitrile is ‘activated’ (protonated) by the acidic catalyst ETT (5-(ethylthio)-1H-tetrazole). Synthesis of peptides can also use activators. Amide bond formation between an amine and carboxylic acid is slow, and as such usually requires “coupling reagents” or activators. Activation of the carboxyl group generally involves the formation of an “active ester” in situ. Activators for peptide synthesis may be Carbodiimides such as dicyclohexylcarbodiimide (DCC) and diisopropylcarbodiimide (DIC). Alternatively, aminium/uronium or phosphonium salt of a non-nucleophilic anion (tetrafluoroborate or hexafluorophosphate) may be used as activators for peptide synthesis. Examples of aminium/uronium reagents include HATU (HOAt), HBTU/TBTU (HOBt) and HCTU (6-ClHOBt). HBTU and TBTU differ only in the choice of anion. Phosphonium reagents include PyBOP (HOBt) and PyAOP (HOAt).

At **808**, an additional monomer is coupled to the growing polymer. Once the protecting group is removed, the polymer chain is available for addition of the next monomer. Any type of polymerization technique, such as by chemical or enzymatic methods, may be used to add an additional monomer on to the final monomer on the growing polymer chain (e.g., the terminal monomer **112** shown in FIG. 1). Reactive groups on any or all of the monomers in the polymer chain may be protected by additional protecting groups that have a different cleavage mechanism than the protecting group that prevents extension of the polymer chain. The additional monomer added onto the growing polymer chain may also include a protecting group attached by a chain linker. Thus, the protecting group prevents multiple incorporation of the same monomer onto the growing polymer chain.

Process **800** may proceed iteratively from **808** and return to **804** where the protecting group is removed by cleavage of a linker and polymer extension continues.

At **810**, upon completion of synthesis of the polymer, the polymer chain may be separated from the substrate by cleaving a surface linker using any of the selectively-controllable or conventional cleavage techniques described in this disclosure. Prior to separation from the substrate or following separation from the substrate, protecting groups on any reactive groups of the monomers of the polymer chain may be removed.

If separation of the polymer chain from the substrate is implemented by cleavage of selectively-cleavable linkers such as the surface linker **104**, the surface linker **300**, the dual-purpose linker **400**, or another linker, individual polymers are groups of polymers may be released while synthesis continues for other polymers. Assuming orthogonality between the linkers attaching the polymer chain to the substrate and the linkers attaching a protecting group that prevents further polymerization, continuation of polymerization and release of a polymer from the substrate may be controlled independently.

Furthermore, depending on specific linker chemistry, surface linkers that have been cleaved to release bound polymers may be reused to start synthesis of a new polymer. For example, the surface linkers **204** for group D **214** in FIG. 2 may be reused after selective release of the attached polymer

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chains. This particular while polymer synthesis is ongoing for one or more of the other groups on array **200**.

Illustrative Embodiments

The following clauses described multiple possible embodiments for implementing the features described in this disclosure. The various embodiments described herein are not limiting nor is every feature from any given embodiment required to be present in another embodiment. Any two or more of the embodiments may be combined together unless context clearly indicates otherwise. As used herein in this document “or” means and/or. For example, “A or B” means A without B, B without A, or A and B. As used herein, “comprising” means including all listed features and potentially including addition of other features that are not listed. “Consisting essentially of” means including the listed features and those additional features that do not materially affect the basic and novel characteristics of the listed features. “Consisting of” means only the listed features to the exclusion of any feature not listed.

Clause 1. A method of site-specific cleavage of attached polymers from a substrate, the method comprising: preparing the substrate with selectively-cleavable linkers; growing polymers on the selectively-cleavable linkers; and selectively cleaving a portion of the selectively-cleavable linkers by changing local conditions on a portion of the substrate thereby releasing polymers attached to the portion of the substrate.

Clause 2. The method of clause 1, wherein the selectively-cleavable linkers comprise electrochemically-cleavable linkers and the substrate comprises spatially-addressable electrodes.

Clause 3. The method of any of clauses 1-2, wherein the selectively-cleavable linkers comprise photolabile linkers and the local conditions comprise light contacting the photolabile linkers.

Clause 4. The method of any of clauses 1-3, wherein the selectively-cleavable linkers comprise thermolabile linkers and the substrate comprises spatially-addressable resistors.

Clause 5. The method of any of clauses 1-4, wherein the selectively-cleavable linkers comprise acid-labile linkers or base-labile linkers and selectively cleaving a portion of the selectively-cleavable linkers comprises selectively applying acid or base to the portion of the substrate without applying acid or base to other portions of the substrate.

Clause 6. The method of clause 5, wherein selectively applying acid or base to the portion of the substrate comprises location-specific printing of the acid or the base onto the portion of the substrate.

Clause 7. The method of any of clauses 1-6, wherein the selectively-cleavable linkers are two-stage cleavable linkers comprising a first cleavable group that, when removed, exposes a second cleavable group that attaches the polymers to the substrate.

Clause 8. The method of any of clauses 1-7, wherein the polymers comprise oligonucleotides or polypeptides.

Clause 9. An article of manufacture comprising: a substrate comprising an array of independently-addressable electrodes; a surface linker bound to the substrate in proximity to an electrode of the array, the surface linker comprising at least one chemical bond that cleaves in response to activation of the electrode; and a polymer attached to the surface linker.

Clause 10. The article of manufacture of clause 9, wherein the substrate comprises complementary metal-oxide-semiconductor integrated circuits.

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Clause 11. The article of manufacture of any of clauses 9-10, wherein the at least one chemical bond comprises a bond in an ester connected to an alkoxy group.

Clause 12. The article of manufacture of any of clauses 9-11, wherein the surface linker comprises an amino acid sequence that becomes accessible to a peptidase upon cleavage of the at least one chemical bond.

Clause 13. The article of manufacture of clause 12, wherein the at least one chemical bond comprises a bond between a nitrogen of a peptide bond and an allyl group.

Clause 14. The article of manufacture of any of clauses 9-13, wherein the surface linker comprises a thermolabile linker and the independently addressable electrodes are coupled to resistors configured to generate heat.

Clause 15. The article of manufacture of any of clauses 9-14, wherein the polymer comprises an oligonucleotide having a sequence that encodes digital information.

Clause 16. A selectively-cleavable chain linker comprising: an attachment to a monomer of a polymer chain; a protecting group that prevents extension of the polymer chain or protects a side group of the monomer from reacting; and a bond cleaved by a redox reaction upon addition of electrons.

Clause 17. The selectively-cleavable chain linker of clause 16, wherein the monomer comprises a nucleotide or an amino acid.

Clause 18. The selectively-cleavable chain linker of any of clauses 16-17, wherein the protecting group is selected from the group comprising acetyl, carbamate, benzyl, benzoyl, Boc, and Fmoc.

Clause 19. The selectively-cleavable chain linker of any of clauses 16-18, wherein the protecting group prevents extension of the polymer chain due to steric hindrance.

Clause 20. The selectively-cleavable chain linker of any of clauses 16-19, further comprising a covalently-attached enzyme that polymerizes the monomer into the polymer chain.

Clause 21. A method of controlling addition of nucleosides in oligonucleotide synthesis, the method comprising: selectively cleaving electrochemically-cleavable linkers linked to nucleobases on terminal nucleosides by inducing a cyclization reaction through removal of protecting groups by activation of a portion of electrodes in a microelectrode array, wherein the terminal nucleosides are nucleosides on ends of oligonucleotides attached to the microelectrode array; and coupling nucleosides with nucleobases linked to protecting groups by electrochemically-cleavable linkers to the nucleosides at which the protecting groups were removed.

Clause 22. The method of clause 21, wherein the nucleosides are nucleoside triphosphates.

Clause 23. The method of clause 23, wherein the coupling of the nucleosides is catalyzed by terminal deoxynucleotidyl transferase (TdT).

Clause 24. The method of clause 21, wherein the nucleosides are nucleoside phosphoramidites.

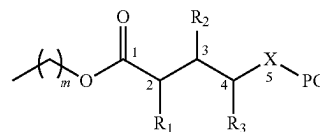
Clause 25. The method of clause 24, wherein the coupling of the nucleosides adds an additional phosphoramidite.

Clause 26. The method of any of clauses 21-25, wherein the electrochemically-cleavable linkers comprise an amide linkage and an amide that forms a lactam when the protecting groups are removed.

Clause 27. The method of any of clauses 21-25, wherein the electrochemically-cleavable linkers comprise an ester linkage and a protected alcohol that forms a lactone when the protecting groups are removed.

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Clause 28. The method of any of clauses 21-27, wherein the electrochemically-cleavable linkers have the structure:



wherein m represents one or more carbon atoms, R_1 , R_2 , and R_3 are each independently a saturated alkyl with 1 to 10 carbon atoms, X is oxygen (O) or an amine (NH), and PG is a protecting group.

Clause 29. The method of clause 28, wherein terminal deoxynucleotidyl transferase (TdT) is attached to R_1 , R_2 , R_3 , or PG.

Clause 30. The method of any of clauses 21-29, wherein the protecting groups are one of benzonitrile, dimethoxytrityl (DMT), tert-butyloxycarbonyl (Boc), benzyl, or N-benzyloxycarbonyl.

Clause 31. The method of any of clauses 21-30, wherein the protecting groups are removed by directly mediated electrochemical cleavage.

Clause 32. The method of any of clauses 21-30, wherein the protecting groups are removed by indirectly mediated electrochemical cleavage.

Clause 33. The method of any of clauses 21-32, wherein the microelectrode array comprises complementary metal-oxide-semiconductor (CMOS) integrated circuits.

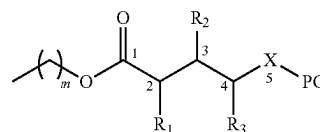
Clause 34. The method of any of clauses 21-33, further comprising repeating the cleaving and coupling over multiple cycles of nucleoside addition where the portion of the electrodes in the microelectrode array that are activated is changed each cycle.

Clause 35. The method of any of clauses 21-34, wherein the nucleosides are nucleoside triphosphates, the electrochemically-cleavable linkers are attached to TdT and the TdT that catalyzes the oligonucleotide synthesis is the TdT attached to the electrochemically-cleavable linkers.

Clause 36. A nucleoside with a nucleobase linked to a protecting group by an electrochemically-cleavable linker, wherein the electrochemically-cleavable linker comprises (i) an ester linkage and a protected alcohol that forms a lactone when the protecting group is removed or (ii) an amide linkage and amine that forms a lactam when the protecting group is removed.

Clause 37. The nucleoside triphosphate of clause 36, wherein the protecting group is one of benzonitrile, dimethoxytrityl (DMT), tert-butyloxycarbonyl (Boc), benzyl, or N-benzyloxycarbonyl.

Clause 38. The nucleoside triphosphate of clause 36 or 37, wherein the electrochemically-cleavable linker has the structure:



wherein m represents one or more carbon atoms, R_1 , R_2 , and R_3 are each independently a saturated alkyl with 1 to 10 carbon atoms, X is oxygen (O) or an amine (NH), and PG is a protecting group.

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Clause 39. The nucleoside triphosphate of clause 38, wherein a DNA polymerase is attached to R₁, R₂, R₃, or PG.

Clause 40. The nucleoside triphosphate of clause 39, wherein the DNA polymerase is terminal deoxynucleotidyl transferase (TdT).

CONCLUSION

Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts are disclosed as example forms of implementing the claims.

The terms “a,” “an,” “the” and similar referents used in the context of describing the invention are to be construed to cover both the singular and the plural unless otherwise indicated herein or clearly contradicted by context. The terms “based on,” “based upon,” and similar referents are to be construed as meaning “based at least in part” which includes being “based in part” and “based in whole,” unless otherwise indicated or clearly contradicted by context. The terms “portion,” “part,” or similar referents are to be construed as meaning at least a portion or part of the whole including up to the entire noun referenced. As used herein, “approximately” or “about” or similar referents denote a range of $\pm 10\%$ of the stated value.

Certain embodiments are described herein, including the best mode known to the inventors for carrying out the invention. Of course, variations on these described embodiments will become apparent to those of ordinary skill in the art upon reading the foregoing description. Skilled artisans will know how to employ such variations as appropriate, and the embodiments disclosed herein may be practiced otherwise than specifically described. Accordingly, all modifications and equivalents of the subject matter recited in the claims appended hereto are included within the scope of this disclosure. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the invention unless otherwise indicated herein or otherwise clearly contradicted by context.

Furthermore, references have been made to publications, patents and/or patent applications throughout this specification. Each of the cited references is individually incorporated herein by reference for its particular cited teachings as well as for all that it discloses.

The invention claimed is:

1. A method of controlling addition of nucleosides in oligonucleotide synthesis, the method comprising:

selectively cleaving electrochemically-cleavable linkers linked to nucleobases on terminal nucleosides by inducing a cyclization reaction through removal of protecting groups by activation of a portion of electrodes in a microelectrode array, wherein the protecting groups are removed by directly mediated electrochemical cleavage and wherein the terminal nucleosides are nucleosides on ends of oligonucleotides attached to the microelectrode array; and

coupling free nucleosides with nucleobases linked to protecting groups by electrochemically-cleavable linkers to the terminal nucleosides at which the protecting groups were removed.

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2. The method of claim 1, wherein the free nucleosides are nucleoside triphosphates.

3. The method of claim 2, wherein the coupling of the free nucleosides is catalyzed by terminal deoxynucleotidyl transferase (TdT).

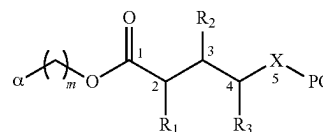
4. The method of claim 1, wherein the free nucleosides are nucleoside phosphoramidites.

5. The method of claim 4, wherein the coupling of the free nucleosides adds an additional phosphoramidite.

6. The method of claim 1, wherein the electrochemically-cleavable linkers comprise an amide linkage and an amide that forms a lactam when the protecting groups are removed.

7. The method of claim 1, wherein the electrochemically-cleavable linkers comprise an ester linkage and a protected alcohol that forms a lactone when the protecting groups are removed.

8. The method of claim 1, wherein the electrochemically-cleavable linkers have the structure:



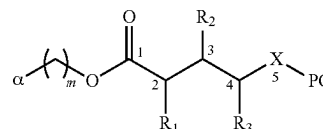
wherein α represents a point of connection to a nucleobase on the terminal nucleosides, m represents one or more carbon atoms, R₁, R₂, and R₃ are each independently a saturated alkyl with 1 to 10 carbon atoms or one of R₁, R₂, and R₃ is a DNA polymerase, X is oxygen (O) or an amine (NH), and PG is a protecting group or the DNA polymerase.

9. The method of claim 8, wherein the DNA polymerase is attached to R₁, R₂, R₃, or PG.

10. The method of claim 8, wherein the DNA polymerase is terminal deoxynucleotidyl transferase (TdT).

11. The method of claim 8, wherein the DNA polymerase is attached to one of R₁, R₂, R₃ and PG is one of benzonitrile, dimethoxytrityl (DMT), tert-butyloxycarbonyl (Boc), benzyl, or N-benzyloxycarbonyl.

12. A nucleoside with a nucleobase linked to a protecting group by an electrochemically-cleavable linker, wherein the electrochemically-cleavable linker comprises (i) an ester linkage and a protected alcohol that forms a lactone when the protecting group is removed or (ii) an amide linkage and amine that forms a lactam when the protecting group is removed, wherein the electrochemically-cleavable linker has the structure:



wherein α represents a point of connection to the nucleobase on terminal nucleosides, m represents one or more carbon atoms, R₁, R₂, and R₃ are each independently a saturated alkyl with 1 to 10 carbon atoms or one of R₁, R₂, and R₃ is a DNA polymerase, X is oxygen (O) or an amine (NH), and PG is a protecting group or the DNA polymerase.

13. The method of claim 1, wherein the protecting groups are one of benzonitrile, dimethoxytrityl (DMT), tert-butyloxycarbonyl (Boc), benzyl, or N-benzyloxycarbonyl.

14. The method of claim 1, wherein the microelectrode array comprises complementary metal-oxide-semiconductor (CMOS) integrated circuits.

15. The method of claim 1, further comprising repeating the cleaving and coupling over multiple cycles of nucleoside addition where the portion of the electrodes in the micro- 5 electrode array that are activated is changed each cycle.

16. The method of claim 1, wherein the nucleosides are nucleoside triphosphates, the electrochemically-cleavable linkers are attached to TdT and the TdT that catalyzes the 10 oligonucleotide synthesis is the TdT attached to the electrochemically-cleavable linkers.

17. The nucleoside triphosphate of claim 12, wherein the DNA polymerase is attached to one of R₁, R₂, R₃ and PG is one of benzonitrile, dimethoxytrityl (DMT), tert-butyloxy- 15 carbonyl (Boc), benzyl, or N-benzyloxycarbonyl.

18. The nucleoside triphosphate of claim 12, wherein the protecting group is one of benzonitrile, dimethoxytrityl (DMT), tert-butyloxycarbonyl (Boc), benzyl, or N-benzy- 20 loxycarbonyl.

19. The nucleoside triphosphate of claim 12, wherein a DNA polymerase is attached to R₁, R₂, R₃, or PG.

20. The nucleoside triphosphate of claim 19, wherein the DNA polymerase is terminal deoxynucleotidyl transferase (TdT). 25

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